AI ENGINEER / AGENTS & AUTOMATION

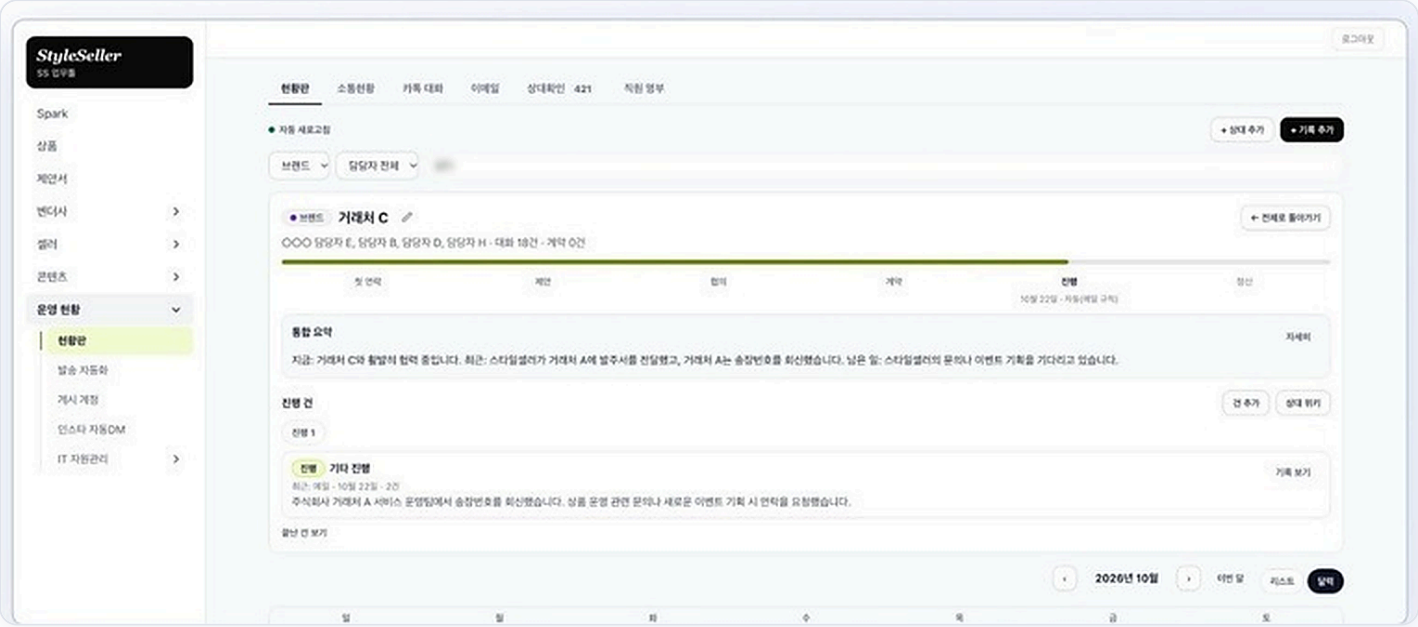
Portfolio

Turning everyday business work into AI systems people use.

Jinhee Mok

AI & Automation Engineer · Design, deployment and operations

At StyleSeller, I build tools for partner support, shared knowledge and day-to-day operations. I deploy them to real staff PCs, investigate failures and improve the workflows after launch.



Production dashboard. Identifying details are blurred.

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One place to follow a business partner's conversations, orders and deal stage.

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Reliable pricing, reviewable catalog candidates and controlled personalized KakaoTalk sends.

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Staff PC Deployment

Shipping and operating tools on staff Macs and Windows PCs, then staying responsible after installation.

Selected work for Forward Deployed Software Engineer roles.

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Resume

Summary

I build AI agents and automation for repetitive work, and stay involved after launch. Since joining StyleSeller in November 2025, I have delivered tools for partner support, email replies, knowledge search and staff PCs. Before software, I worked in daycare operations and English teaching. I use Claude Code for implementation while defining the scope and reviewing the results myself.

Experience

StyleSeller · AI & Automation Project Lead (PL)

2025.11–present

- Designed, built and operated partner-support agents, Spark and work automation.
- Built a shared knowledge hub and claim-by-claim checks for email drafts.
- Skill-router evaluation: 119/132 correct (90.15%), re-measured after fixing answer-key errors and a fake skill.
- Automated KakaoTalk and email collection, supplier catalog registration, pricing and partner matching.
- Deployed tools to eleven staff PCs with one-line install and auto-update (the Windows collector updates by reinstall); recovered a cloud-DB outage that stopped the work-tool API the same day and moved 4 databases to self-hosting; operated GCP VMs, Docker and NGINX.

We:Review · AI Development Intern

2025.08–2025.10

Built an OCR-to-structured-data receipt pipeline with PaddleOCR, DBSCAN, Qwen2.5-7B and FastAPI. Took part in deployment: review accuracy 70% → 95%, review time reduced by 85%.

Jeongsang · Pots&Pans · Montekids · English Teacher

2018.08–2024.02

Designed lessons for ages 5–15, prepared materials and individual learning plans, and held parent consultations.

Jeonwon Daycare / After-School · Operations Manager

2015.03–2018.02

Hired teachers, planned the curriculum and events, worked with parents and the steering committee, and managed daily operations as acting director.

Education

Cheongju University · Law · 2008–2012

Academic Credit Bank System · Business Administration · 2021–2023

Training

Himedia · Full-Stack (960h) · 2024.04–10

Himedia · Advanced AI · 2024.10–12

Korea IT Business Promotion Association · Generative AI · 2025.05–07

Skills

AI/LLM · Gemini, LiteLLM, LightRAG, MCP, tool calling, evaluation design

Backend · Python, FastAPI, TypeScript, Node.js, PostgreSQL

Infrastructure · GCP, Docker, NGINX, Vercel, GitHub Actions, n8n

At a Glance

Seven systems I built and operate at StyleSeller, plus one follow-up improvement. Each case explains the problem, the implementation, the choices I made and the limits that remain. Figures carry their measurement dates.

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Rolled out a cheaper skill router and corrected the evaluation. 119/132 correct; estimated 90% lower cost from reducing Gemini input.

LightRAG Knowledge Hub · Email Agent
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Shared search with evidence checks before email drafts reach the assigned staff member.

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Reliable pricing, reviewable catalog candidates and controlled personalized KakaoTalk sends.

Staff PC Deployment
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Shipping and operating tools on staff Macs and Windows PCs, then staying responsible after installation.

Terms used in this portfolio

- **Group buy** · a time-limited sale in which an influencer sells a brand's products to followers
- **Seller** · a social-media influencer who runs group buys
- **Brand (supplier)** · the company that supplies the products
- **Sales partner** · a company that runs group buys with its own sellers; labeled "vendor" in the work tool
- **Vendor / business partner** · in the vendor support case, brands and sales partners together
- **Work tool** · StyleSeller's internal web app for pricing, catalog, matching and sends
- **Spark** · staff assistant that answers read-only questions over four business databases and hands work to agents
- **CEO agent** · the top-level agent that writes work orders; not a person
- **Paperclip · Hermes** · an open-source agent issue manager I forked · the agent runtime
- **Jev · BM25** · a judgment model (TypeSafe) that scores skill fit · keyword search

Status Board · Operations Ledger

One place to follow a business partner's conversations, orders and deal stage.

Period	Built October 19–23, 2026 (5 days) · In production
Role	I designed the ledger and matching rules, built the intake API, email collector and six UI tabs, and verified the database behavior and production screens. I used Claude Code for implementation, while setting the scope and reviewing the results myself.
Stack	React 19.2 · Tailwind 4 · Vitest 4 · Python 3.12 · Pydantic v2 · PostgreSQL RLS · Gemini Flash (LiteLLM)

Background

Staff followed the same deal across KakaoTalk rooms, email, Slack and emailed order sheets. I built a dashboard that puts these records on a per-partner calendar and shows deal stages across hundreds of partners.

How it works

1. Collect: KakaoTalk runs three times a day; email every five minutes; Slack uses a connector. Source records enter a local queue before delivery.
2. Prepare: Code filters newsletters and masks buyer details in order-sheet columns. The LLM extracts summaries; cited phrases are checked against the source.
3. Link and store: An idempotent ledger stores each record once. Ordered rules link partners by thread, registration number, room and contact details. Conflicts remain unclassified.
4. Review and follow up: Staff resolve unclear matches and stages. An 18:05 batch writes 3–6-line summaries for up to around 500 partners, stopping at its daily cost limit of KRW 2,000.

> **Scale and evidence.** Communication Status share by stage: first contact 13.1%, proposal 33.8%, negotiation 2.6%, contract 4.5%, in progress 46.0%, settlement 0% (2026-10-24) · Roughly 100–200 records per weekday, hundreds awaiting business partner matching (2026-10-24) · About 150 commits in 2 repositories

Incidents and fixes

All 107 summaries rejected (2026-10-23). The application role lacked execute permission on the summary-saving function. All 107 saves failed with db_error. Granting execute access to that function restored the path.

Too many 'in progress' proposals in the rehearsal (2026-10-23). Broad words such as shipping and inventory triggered false deal signals. I kept only actual deal terms and removed negative phrases such as sales halted and order inquiry before matching.

Shared chatbot shown as the contact person (2026-10-23). A shared chatbot appeared among the five most active room speakers and became the contact for 64 partners. I now use company email accounts, falling back to the account-profile contact.

Technical highlights

1. Retries preserve staff edits. An upsert on (source, source_ref) ignores unchanged content and protects staff-edited records. Duplicate intake does not bump sequence numbers or timestamps.

2. Authorship comes from the source. The server rejects the caller's author value. Database triggers set the staff-edit flag; hidden records cannot be restored by an ordinary update.

Stage rules are enabled one at a time; uncalibrated model confidence never decides a stage. Python, TypeScript and SQL share normalization examples; CI tests content with PGlite as the real app role; UI tests run separately.

Partner linking uses ordered rules, not an LLM; staff resolve key conflicts. Stage updates lock in intake order (partner, then proposal), so they cannot deadlock. Next time I would test each new DB function as the app role; it would have caught the 107 rejections.

> The six tabs, three intake channels and daily summary batch were checked on October 24. Intake was about 100–200 records per weekday. Staff acceptance of suggested stages has not been measured; the server currently trusts the intake-side evidence checks.

Status Board · Operations Ledger

Architecture

Rule (code) · LLM · Staff judgment · DB

1 Collect

KakaoTalk room summaries
3x/day: 09:15, 13:00, 17:55
vendor rooms only, hashed by room

↓

Gmail order inbox
history cursor every 5 min
moves only after the ledger has it all

↓

Slack · general connector
via Hermes · n8n
same intake format, one record at a time

2 Clean & summarize

Rule filter
newsletters and auto-mail dropped before any LLM call

↓

PII masking
order sheets: recipient and orderer columns only
names beside a corporate tag are kept

↓

LLM summary & hints
Gemini Flash family via LiteLLM, temperature 0
business partner candidate, stage hint, deal terms

↓

Source check (at intake)
cited phrase must be an exact substring of the masked text
term values: number, unit, and keyword must all match
mismatches are dropped

3 Into the ledger

Intake API
idempotent upsert on (source, source_ref)
batches of 100 / 256KB, with errors handled for each record

↓

Partner matching
thread → business reg. no. → KakaoTalk room → email → phone → domain → code
conflicting keys: unclassified
new business partners: lock, then look again

↓

Ledger DB (RLS)
The source determines the recorded author.
Changes are added to the history without overwriting earlier entries.

↓

In-progress deals
KakaoTalk contract + mail signals, grouped by business partner, target, round

4 Human confirmation

Partner check
hundreds unclassified
Linking one record connects the whole room or thread. Later records use the same match.

↓

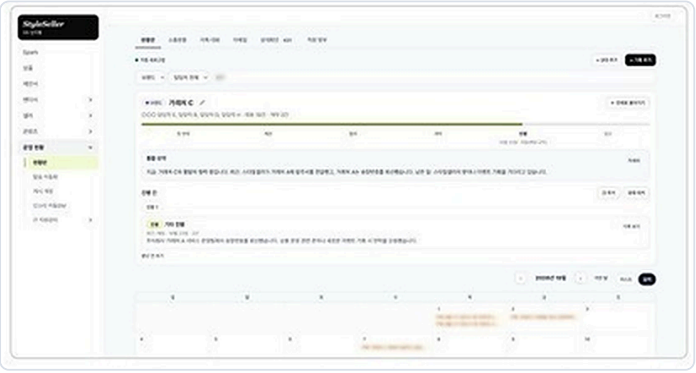
Not a business partner: hide
future records hidden too, reversible

↓

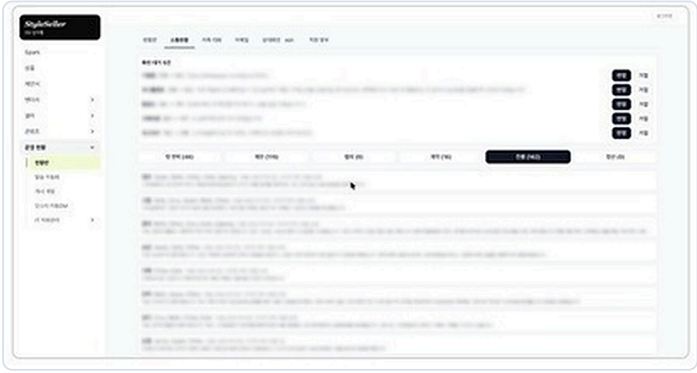
Summary batch, 18:05
up to around 500 business partners, latest 40 records for each partner that day
now / recent / remaining, 3–6 lines
a late retry can't overwrite a newer one

↓

Awaiting review: apply / reject
only a (source, stage) rule match auto-applies
The staff member who applies a change is recorded by email.
rejected stages never return



Partner calendar · conversations, email and order sheets in date order.



Deal-stage board · staff apply or reject pending suggestions.

Technical highlights

3. Conflicting keys stay unclassified. Matching follows a fixed order. Shared public domains are excluded; creating a partner takes a lock and rechecks to avoid duplicates.

4. Extraction needs source evidence. Cited phrases must occur in the masked source. Deal numbers, units and keywords must match; buyer columns are masked while product details remain.

5. Costs and cursors survive concurrency. Cost checking and reservation share one SQL statement. The email cursor advances only after delivery succeeds; repeated failures are quarantined and reported.

Reels

Content Autopilot

StyleSeller · Private repositories

A Reel workflow with predictable stages, a reviewable preview and controlled publishing.

Period	From October 3, 2026, after design approval · Render path verified; publishing staged
Role	I designed the state machine and publishing defaults, built the orchestrator, database publishing slots and QA gate, and checked the production render path and restart behavior. I used AI coding tools; I set the scope and verified results.
Stack	Paperclip sub-issues · LLM-free state machine · Hermes · Gemini Flash (LiteLLM) · image-service · TTS · video renderer · HyperFrames · ffprobe · Google Drive · Blotato · Supabase

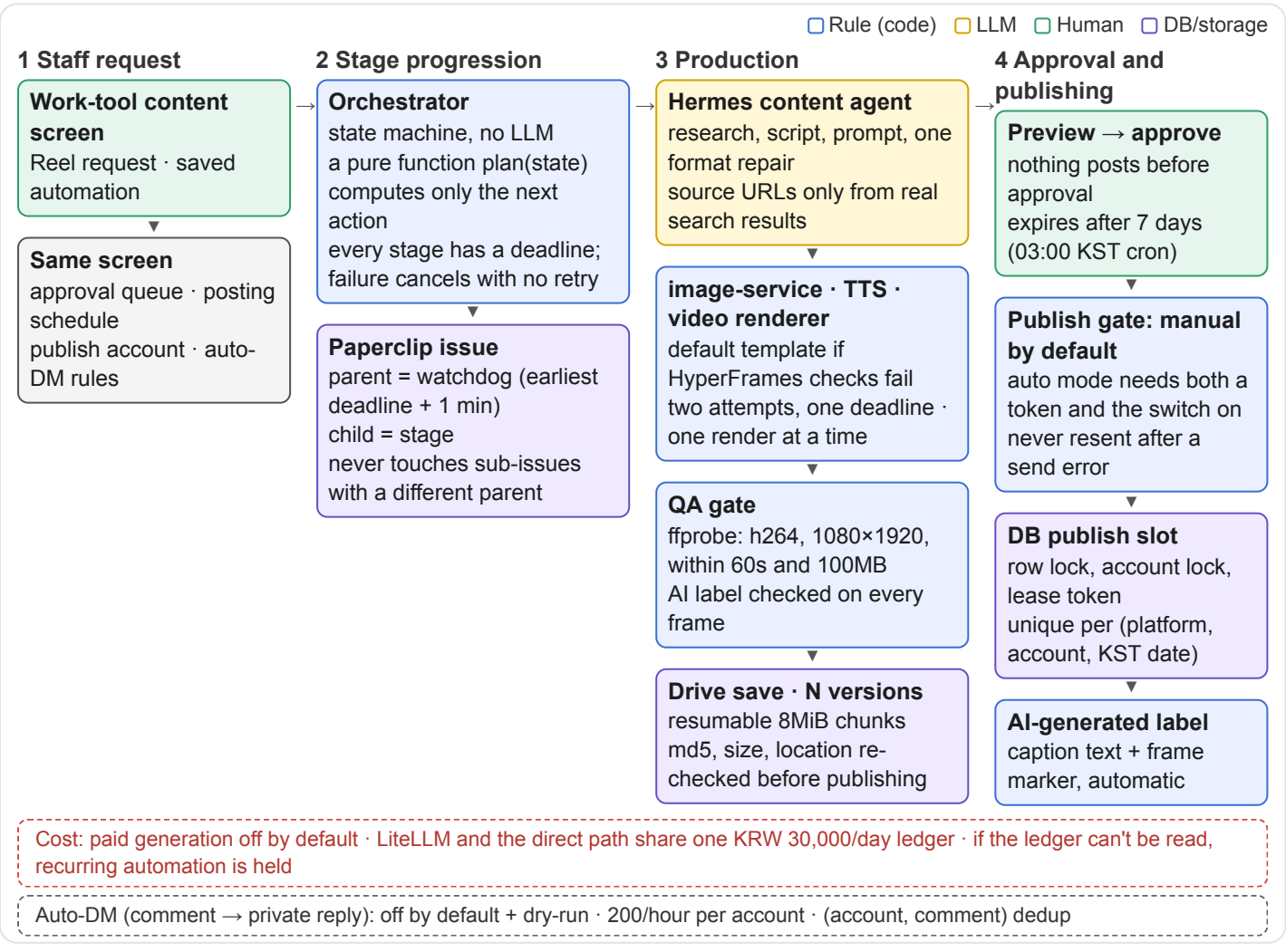
Background

Generating a video costs money, and publishing makes it public. In my workflow, staff request a Reel, inspect the render and approve it in the work tool. Code controls the stages; an agent prepares research, scripts and prompts.

How it works

1. Request: Staff submit a Reel request or start a saved automation. Recurring automation simulates execution by default; paid generation is off.
2. Plan and produce: A state machine selects the next action. Hermes prepares research and a script; the renderer produces a vertical MP4 from a template.
3. Check and approve: QA checks format, dimensions, duration, size and AI markers; a checksum pins the saved file. Staff approve the preview; approval expires after seven days.
4. Publish: Publishing is manual by default. Automated publishing requires a token and switch, a database slot and a final file check. Unknown outcomes are reconciled before another attempt.

Architecture



Scale and evidence. About 122k lines across 3 services, 617 test files (orchestrator 135, Hermes agent 396, video renderer 86) · Production canary verified through saving a 1080×1920 MP4 to Drive (2026-10) · 0 live SNS posts from the new production path. One post per account per day is enforced by the DB; the publish gate caps all accounts at 10 posts per day (2026-10-24)

Defaults and controls

Posting. The default is manual. Auto-posting turns on only when both the token and the switch are present. A partial unique index on (platform, account, KST date) blocks more than 1 post per account per day. If duplicates already exist, the migration stops first.

Paid generation and recurring automations. The paid backend defaults to False. Recurring automations default to dry-run and are released only when the value is exactly "false". Before starting, they read the company's KRW 30,000-per-day ledger, and hold if it can't be read. The cap is clamped with min() so env can't raise it.

Instagram auto-DM. By default the enable switch is off and dry-run is on. The limit is 200 per account per hour, and invalid config values are read toward the safe side. A unique index on (account, comment) stops a single comment from getting two DMs.

Technical highlights

1. Code chooses the next stage. A planner returns actions and an executor applies them. Deadlines wake the parent issue; a rerun does not create a second pipeline or touch another parent's work.

2. Repairs are bounded. JSON errors get one repair call with up to 8 errors of 120 characters each. A failed motion plan gets one default-template render within the original deadline; paid or public actions need another human decision.

3. Sources must come from real search. Cited URLs must appear in tool results or the supplied input. Model-written evidence fields are discarded; unsupported search claims fail verification.

4. Unknown publishing outcomes are reconciled. A marker is written before the POST. Uncertain results stay pending; saved submission IDs are checked after restart instead of publishing again.

5. The database allocates publishing slots. Row and account locks issue a 30–900-second lease after checking the cap. A unique index enforces one post per account per day across processes.

6. Cost paths share one ledger. To close a hole where direct calls bypassed the proxy cap, LiteLLM and direct Vertex calls add to the same Redis key. If the ledger cannot be read, individual calls pass but recurring automation starts are held. Reads use a Redis user without write access.

7. Approval is tied to the approved file. Before posting, recheck the Drive file type, size, checksum and location. QA checks H.264, 1080×1920, 60 seconds, 100MB and required AI markers. Uploads use 8MiB chunks and resume on HTTP 308.

The production trial generated a 1080×1920 MP4 and saved it to Drive. As of October 24, the new generation path had published no live posts, so it had no measured reach results. Existing videos already publish live through a separate path that swaps only captions and hashtags per account and posts to several accounts at once. The new production path is technically complete, but details such as its prompts are still being refined. Product-price and bundle checks still rely on prompts and human preview; I would add database checks before expanding rollout.

Design choices

- A state machine with no LLM drives stage progression. The LLM only writes research, scripts, and prompts. If the model picked the next stage, every rerun would give a different result.
- Apart from one format repair and one re-render with the default template, it closes on failure instead of retrying automatically. Anything that costs money or goes public, a human clicks again.
- Posting defaults to manual, and the daily cap lives in a DB index, not in app config.

A daily cleanup rejects expired approvals, cancels their issues and closes the rows even if cancellation fails. Registering a new Instagram token still requires developer setup.

Vendor CS Agent

StyleSeller · Private repositories

Tagged KakaoTalk questions answered against partner records, with delivery checked on the staff Mac.

Period	June 2026 onward · In production
Role	I designed the intake and partner gates, built the agent and local send path, added evidence checks and delivery confirmation, and investigated failures on real Macs and Windows PCs. I used AI coding tools; I set the scope and verified results.
Stack	macOS kakaocli fork (Swift) · read-only Windows collector (Python) · Python 3.12 · Gemini Flash (LiteLLM) · LightRAG 1.5.6 · Sheets/Drive

Background

Partner conversations live on staff computers, but the agent needs shared records to answer consistently. I connected local collection, a server ledger and the knowledge hub, and send replies through a staff Mac. Since September 2, the agent answers tagged questions without follow-up questions or a human handoff step. Code checks amounts, percentages and brands against retrieved evidence before delivery.

How it works

1. On Mac, kakaocli reads only what comes after the last logId in the KakaoTalk DB. On Windows, a read-only collector uploads it. It moves on only after the server answers with 2xx.
2. The server drops the same message if it arrives again within 15 seconds. It stores records in per-partner ledgers, Sheets, and Drive, and twice a day (09:30, 18:10) compiles them into a wiki that goes into the knowledge hub.
3. The business partner ID or name in the room name is checked against the work tool's list of business partners to serve. If no business partner is found on the list, it does not answer.
4. It accepts only questions tagged @company_AI. The regex only detects the tag; the model does the search and writes the answer. Other business partners' names are masked as [other vendor].
5. Amounts, percentages, and brands in the draft are checked against the evidence, and on a mismatch it searches again and rewrites. The reply is sent from a staff Mac, and delivery is confirmed by whether the KakaoTalk DB count went up.

Incidents and fixes

A regex blocked questions before model assessment (until 2026-08-30). Some messages were silently blocked without any LLM judgment. The regex meant as a trigger was also doing the blocking. I cut the regex down to tag detection only and moved the decision on whether to answer to the model.

Background KakaoTalk calls hung (2026-08-17–18). On macOS 15, when the KakaoTalk window was in the background, Accessibility (AX) calls ignored their timeout and hung. 31 of 147 sends (21%) failed permanently. I moved the calls to a separate thread and put a 3-second deadline on them with a semaphore (past that, the wait is cut off). Since 09-08 it also re-checks delivery after sending.

Flood of past conversations (2026-09-16). Three Windows PCs uploaded 2,500 records in 50 minutes, the Sheets API was blocked with 429, and rows for the 2 people installed later stayed empty for over an hour. An old collector build had uploaded entire past conversations. I made the server drop records older than a cutoff date and answer 200. Answering with 5xx makes the collector upload the same records again.

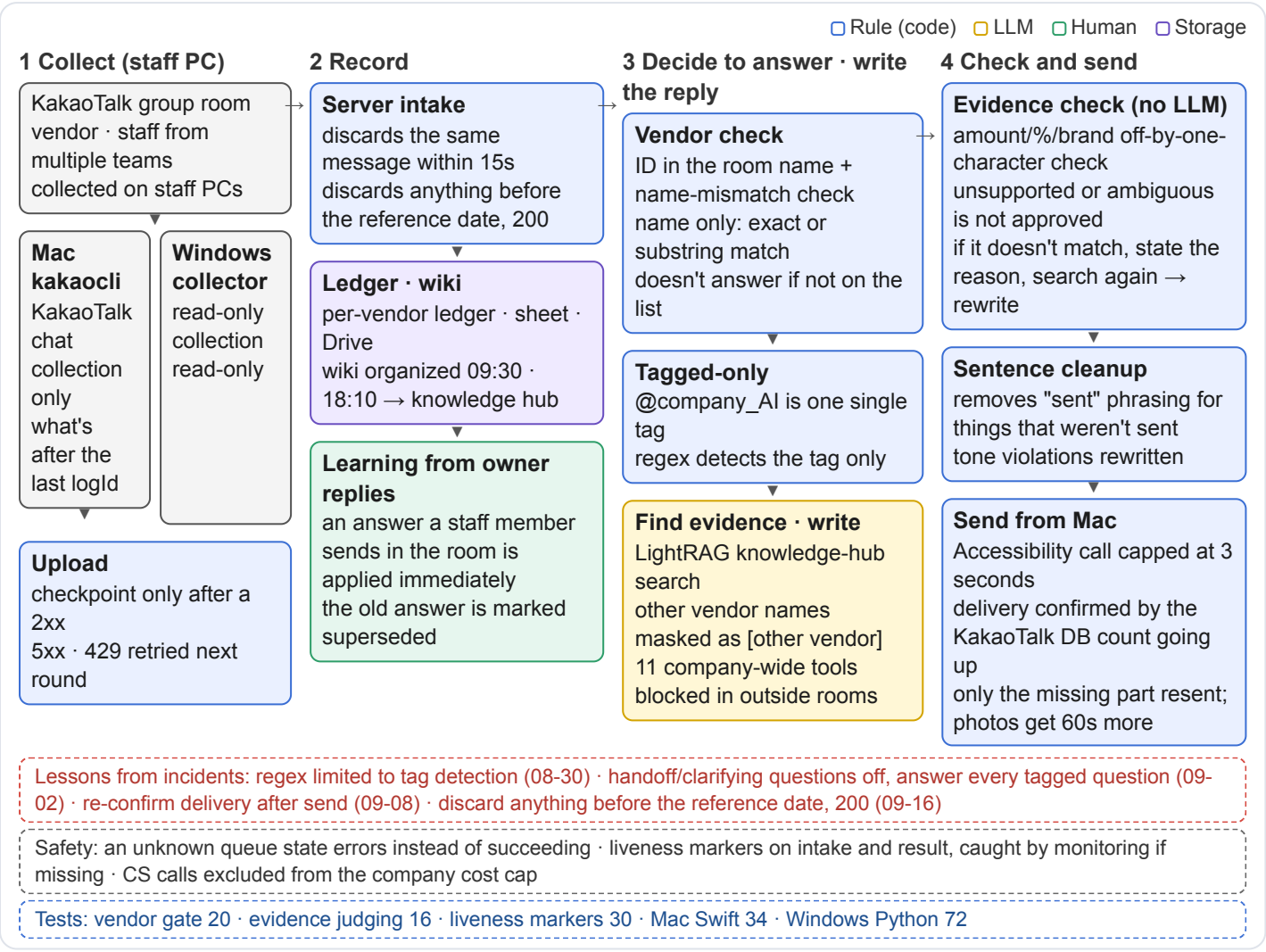
> **Scale and evidence.** Vendor agent code, tests, and config: 61K lines total, 600 commits, 107 test files (2026-09-23) · CS Copilot extension and local sender: 631 commits, 20,596 lines, 56 tests (2026-09-23) · Tests: business partner gate 20, evidence judgment 16, liveness markers 30, Mac Swift 34, Windows Python 72

Direct staff replies are learned at once; the older answer to the same question is marked superseded.

> Processing writes start and result markers so monitoring can detect a tagged question that never reaches an outcome. The Windows collector still updates by reinstalling. Vendor and influencer support calls are excluded from the company daily cost cap. I would include a collector updater and server cutoff date from the first release.

Vendor CS Agent

Architecture



Technical highlights

1. Require a clear partner match. The "(1234)" prefix in a room name is read as the ID. If the name part points better to another vendor at 0.80+ jamo similarity, it is blocked as an ID typo; similarity only blocks, never links. Without an ID, only one exact or substring name match counts. Common suffixes alone scored 0.81–0.88, so fuzzy linking was removed; ambiguous rooms get no answer.

2. Deduplicate without losing room changes. kakaocli re-sends unread rooms' latest message every 2 seconds; one message arrived 51 times (170–200 messages/min). A 15-second key of room, message, unread count and direction drops repeats; the unread count lets a reopened room pass.

3. Check numbers and brands in code. Normalize amounts and percentages (for example, Korean shorthand for 30,000 won becomes 30000 KRW), compare them with evidence and flag near-matching brands. Missing or ambiguous evidence triggers another search and rewrite.

4. Do not claim an action that never ran. Remove completed-send wording when no send occurred. Retries read the real queue state; only failed items are re-queued.

5. Keep external rooms within scope. Other partners' names are masked. Eleven company-wide Spark tools are blocked, and the server supplies the partner identity for product lookup.

6. Confirm delivery locally. Count new text and photos in the KakaoTalk database. Retry only missing parts, allow extra time for photos, then report completion. In a 70-person room, a duplicate photo is worse than a missing one.

7. Acknowledge and monitor the processing path. Uploads skip rejected 422/413 items; 429/5xx stop the batch for retry. Start and result markers detect tagged questions that never reach an outcome; this path has 30 tests. The earlier human-handoff branch remains disabled and covered by tests.

Spark

Agent Platform · Spark

StyleSeller · Private repositories

Work orders, agent delegation and tool execution with controls enforced by the server.

Period	Platform from May 2026 (started May 17) · Spark and Work OS designed July 18–19
Role	I designed work orders and delegation limits, integrated shared skills and tools, built single-use approvals and cost controls, and wrote evaluation cases and deployment checks. I used AI coding tools; I set the scope and verified results.
Stack	Paperclip fork (Express 5, Drizzle, PostgreSQL) · MCP SDK 1.29 · Hermes · Python 3.12 · FastAPI · LiteLLM · Redis · Langfuse 4.50 · Gemini Flash (LiteLLM)

Background

An agent can change the meaning of a request as it delegates work, and a tool can write data, spend money or publish content. I forked Paperclip and added Spark, Work OS and company services, with execution controls in server code. Staff use Spark to query four business databases and hand work to the CEO agent.

How it works

1. Work requests from Spark, Slack and the work tool go to the CEO agent. Only structured requests from the work tool go straight to the content orchestrator, and they still leave a work order record.
2. The CEO writes work order v1. It holds the original text as an attachment with its sha256, the goal, completion criteria, constraints, and whether approval is required.
3. An overall orchestrator reporting directly to the CEO (the Hermes dispatcher) assigns it to one of at most 10 department heads, such as Supervisor, campaigns, ads, content, build, review and deploy. The server caps delegation depth at 2 and delegations per work order at 5, and blocks A→B→A bounce-backs. Turning it off falls back to direct delegation by the CEO. Work that goes to the content department head is handed on again to the content orchestrator, a separate service.
4. BM25 search pulls candidates from all shared skills, the judge model Jev checks gating and fit, and Gemini makes the final pick of one skill. If Jev fails or times out, it is skipped and Gemini picks directly from the full list.
5. Tools run through a 6-stage gate and human approval. Publishing goes through the Publisher gateway (a single service dedicated to publishing). The default is manual, and the DB enforces one post per account per day.

Incidents and fixes

Spark shut down entirely (2026-10-24). A new Spark tool was missing its DB registration row, so the boot check failed and every Spark call was rejected. I built it to block when the check fails, and it did exactly that. After recovering with a restart, I added a rule that every tool's registration row ships as a migration, plus a regression test.

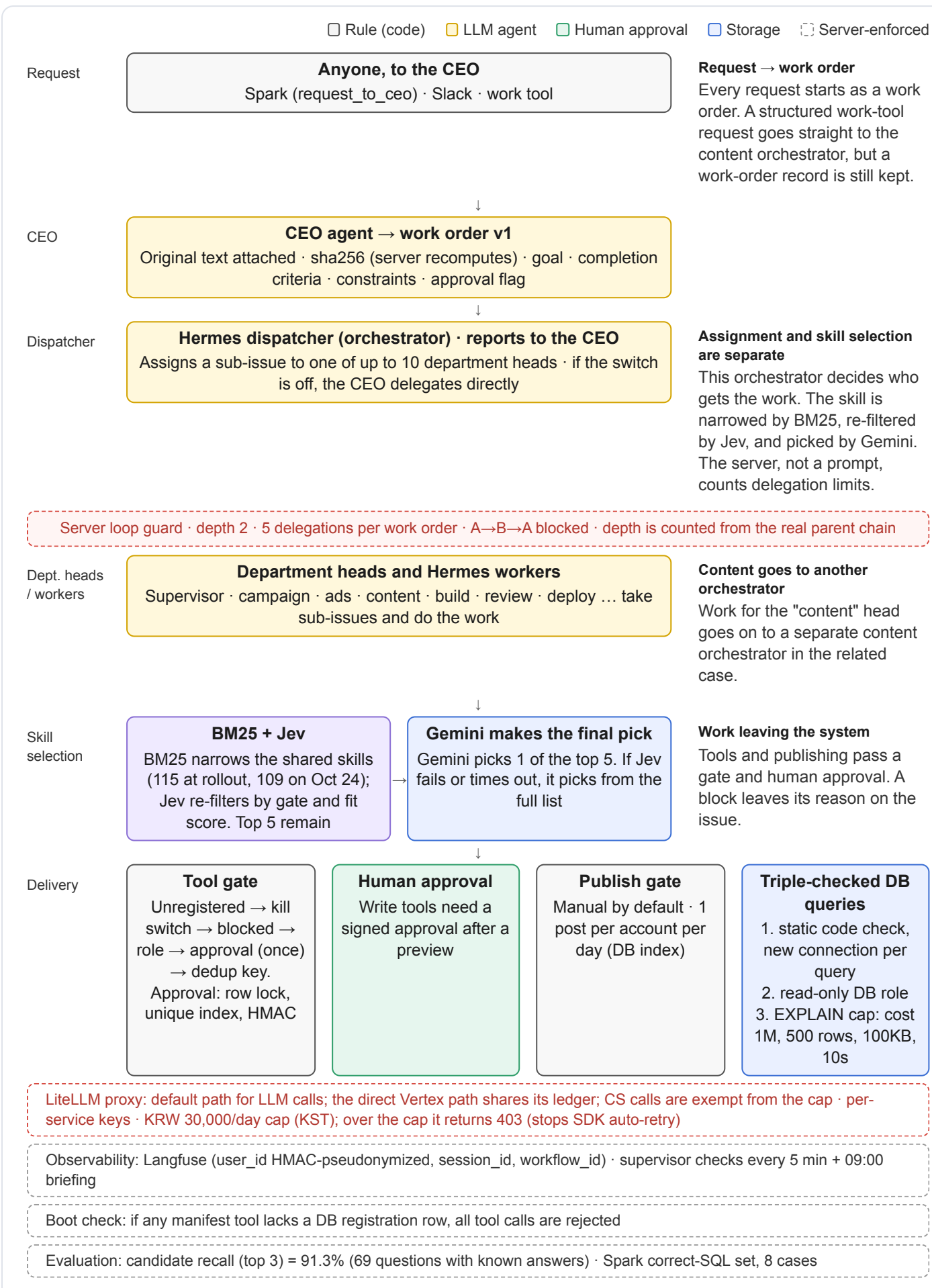
Reclaiming a job mid-render (2026-10-24). The server reaper treated a video job running past 5 minutes as stalled and reclaimed it, and the result that arrived 7 min 40 s later was rejected with 409. I made the job send a heartbeat every 120 seconds and stop once the issue deadline passes. Jobs that are really stuck still get reclaimed. In production I confirmed heartbeats received (200) and 0 reclaims.

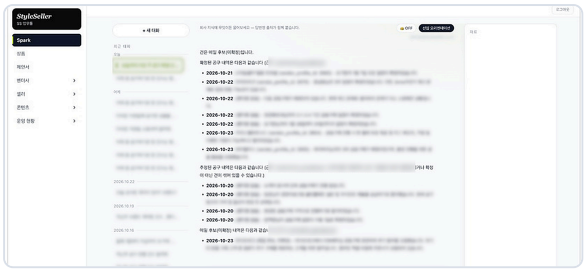
Made the Jev router lightweight (2026-10-23). For speed and a lighter footprint, I moved the Jev router onto a lightweight model. An adapter only matches the response format, so the existing router code stayed unchanged.

> **Scale and evidence.** About 5,000 of my own commits (excluding merges and upstream), about 2,200 test files (2026-10-24) · 109 shared skills (October 24, 2026), 66 trigger eval files (1,282 sentences), 67 tool manifests

After two Jev failures, a 300-second circuit breaker skips that step. Rollout groups use the first eight SHA-256 characters. The server counts delegation depth from the real parent chain, rejecting forged depth and a sixth delegation. Company features attach as HTTP services outside the Paperclip core, except Work OS and Spark. The tool-execution switch is off by default and enabled per tool. Delegation limits live in the server core because plugins only see an issue after it exists. Next time I would compare the tool manifest with registry rows in CI; the 10-24 outage surfaced only at boot.

Architecture





Spark in production. Question titles and partner names are blurred; answers identify their source tool.

Approvals refer to exactly what staff reviewed

A write approval signs the input and preview hashes, requester, run ID and expiry. A row lock and unique index prevent reuse; approval is consumed even when execution fails. A concurrency test confirmed one execution from ten simultaneous requests. Hashing the original request also lets the server detect changes across the delegation chain.

Read-only database access needs several controls

Spark uses a new connection for each query, filters dangerous functions and writing CTEs, rejects EXPLAIN costs above 1,000,000, and limits execution to ten seconds and results to 500 rows or 100KB. Reusing a connection could carry changed session settings into another person's query.

An outage exposed a missing deployment check

On October 24, a new tool lacked its database registration row. The startup check rejected all Spark calls. After recovery, I required registration rows to ship as migrations and added a regression test. A separate long render was reclaimed as stalled; a 120-second heartbeat kept active renders alive while preserving recovery for truly stalled jobs.

Cost and observability

Model calls pass through LiteLLM with service-specific keys. The daily paid cap returns a non-retryable response rather than a rate-limit response that the SDK would retry automatically. Langfuse receives a pseudonymized user ID, session ID and workflow ID. These records let me connect a user request to the tools, model calls and operational checks that handled it.

Measured scope

The October 24 snapshot contained 109 shared skills, 67 tool manifests and 66 trigger-evaluation files. Candidate recall was 85.5% at top one and 91.3% at top three on 69 questions with known answers; median latency was about 1.09 seconds. This measures whether the correct candidate was present, not final answer accuracy. The following Jev case measures final skill selection separately.

Design choices and alternatives

- I chose a full index over per-role skill bundles. On a 57-case selection set both scored the same accuracy, 0.947 (p=1.0). But the bundle approach missed skills that only get used when called by name. The cost difference over 66 cases was 27 KRW.
- The dispatcher assigns work; BM25 shortlists candidates, Jev re-filters them, and Gemini picks the skill. If Jev fails, Gemini still picks directly from the full list, so work does not stop. The worker never picks the skill; Gemini always does.
- Controls sit where they still hold if the model ignores them. The loop guard, single-use approval, and paid cap all live in the server and DB.

Design notes

- The work order is NFC-normalized and locked with its sha256; the server overwrites client hashes. An external benchmark found fewer supervisor errors when the original, not a summary, is passed on.
- Approvals carry an HMAC-SHA256 signature, so database write access alone cannot forge one; the dedup key has no timestamp, so retries add no approvals.

Jev Skill Router Upgrade

Improving skill selection started with checking the evaluation itself.

I designed the test, integrated the router, investigated failures and chose the rollout settings. Hermes uses BM25 to narrow skills, Jev to check fit and Gemini to select the final skill. This narrow-then-judge design follows the same direction as the public SkillRouter study (arXiv 2603.22455). If Jev fails, Gemini selects from the full list. Selection accuracy here is measured separately from the candidate recall on the Spark page.

90.15%
119 of 132 cases correct

115 → 5
Skills reviewed by Gemini

Rolled out
Hermes sub-agents

Corrections and re-measurement

Five of six flagged false positives were answer-key errors. Some apparent gains existed only on questions already used during tuning. A fake skill in the test folder caused another false conclusion; it did not exist in production. I removed it and reran both sets: 90 + 132 = 222 cases.

The skill catalog grew from 109 on October 24 to 115 at test time. The judge alone reached 45–53%; adding the LLM proposer on October 26 brought results into the 80s. Gains on tuned questions initially looked like +4.4 percentage points, versus +1.2 on new questions. After cleaning and re-measuring, those became zero and +1.5, so I stopped treating the tuned-set gain as evidence.

Evaluation stage	Production setting	Comparison setting
60 cases · answer-key errors present	83.3%	91.7%
90 cases · corrected answer key	88.9%	93.3%
Same 90 · fake skill removed	94.4%	94.4%
132 new cases · later reused	90.15%	91.7%

The rows use corrected labels and changing sets. They are not a clean before-and-after accuracy curve. One of the 132 questions equals 0.76 percentage points.

Rollout and cost reduction

The deployed router reduces Gemini review from all 115 skills to five. Published unit prices and the reduced input imply about 90% lower cost than the full-list approach; this is a calculated estimate. Jev is a judgment model (TypeSafe) that returns probabilities rather than text, so it is faster and cheaper than Gemini: 70–500ms published, 360ms measured median. All three production check questions passed before full rollout.

The estimated cost is about 1/10 of the full-list approach, with similar selection accuracy in this evaluation. I compared settings using saved answers, expanded experiments from 20 to 90 to 132 questions and recorded zero HTTP 429 errors. Before rollout I checked all three changed code locations, backed up configuration, restarted the server and verified its health.

I set the rollout criteria before testing: on unseen questions, selection no worse than the full-list approach, no more wrong skill picks, and lower cost. A higher score alone was not enough to enable a setting.

What I held back

A comparison setting gained only 1.5 percentage points and sometimes chose a skill with fit as low as 0.22 despite high confidence. I left it off.

What the result means

The 132 questions were new when introduced, then reused in tuning. Some questions have multiple valid answers. Catalog changes caused 10 label errors (five in each set); automatic label checks remain unbuilt. The result does not guarantee production-wide accuracy.

RAG LightRAG Knowledge Hub · Email Agent

Shared search with evidence checks before email drafts reach the assigned staff member.

Period	March 31, 2026 onward · In production
Role	I designed workspace registration, search planning and evidence checks, built the mail server and draft checkers, and handled rollout, incident investigation and recovery. I used AI coding tools; I set the scope and verified results.
Stack	Python 3.12 · FastAPI · Pydantic v2 · LightRAG 1.5.6 (Postgres/pgvector + Neo4j 5) · RAG-Anything/MinerU · FlashRank · Gemini Flash (LiteLLM) · Gmail · pytest

Background

Vendor support, influencer support, Spark and partner matching need the same business knowledge. I built a shared LightRAG hub with 13 workspaces, each with an owner, channel and sensitivity level. For shared-mailbox inquiries, the email agent prepares a checked draft in the assigned person's Gmail drafts folder; staff send it.

How it works

1. An LLM planner (temperature 0) reads the workspace descriptions and data catalog and builds up to 12 queries. A plan that fails validation is discarded and the rule router takes over.
2. Code pins the search mode (mix, which uses vector and graph together) at each call site. Agent searches use top_k (number of results fetched) 8, and identical queries are cached for 60 seconds.
3. Evidence is judged by rules. Live values such as price, stock, and settlement are forbidden claims. Refund confirmation, contracts, and bulk sends get a human approval flag, and only the missing evidence items are searched once more.
4. Results are grouped by topic using rerank (the step that re-orders results) scores. Within a group, official corrections and the newest documents go first, and output alternates one result per group.
5. A checker and an LLM verifier compare up to 6 claims in the email draft against the evidence. If a claim has no evidence the draft is rewritten, and if the problem remains the draft is discarded.

Incidents and fixes

Missing rerank (2026-06-19~07-23). On 06-18 we switched to agentic RAG (the LLM plans the search). From the next day, the quality of CEO email drafts dropped. Tracing back through the commit history showed the rerank step had been dropped in the switch. I restored it on 07-23.

Search overload (07-10, 08-12). The influencer CS bot took the whole shared connection pool, so even partner-matching searches returned HTTP 500 errors (07-10). top_k 40 searches triggered Neo4j overload warnings (08-12 analysis). I split the connection pool per workspace and capped agent searches at top_k 8.

Silent graph outage (2026-10-04). When Neo4j stopped, the container still looked healthy. Now /readyz sends RETURN 1 to Neo4j with a 2.5-second limit and reports ok or fail. This failure alone does not return 503, so it does not cause a restart loop. Search answers with 503 and Retry-After 30 after 2 attempts.

> **Scale and evidence.** ~1,470 commits · ~47,000 lines of product code · ~33,600 lines of tests (10-24) · 27 routers · ~1,750 test functions · 13 workspaces (10-24) · Search eval set of 18 cases (query, expected workspace, required evidence, forbidden claims)

Sent-mail learning missed 48 of 1,943 messages at rollout. I changed cursor handling to overlapping ten-minute windows, lagging current time by 180 seconds. Sent mail, not AI drafts, feeds the knowledge hub. Only server images that pass pytest are deployed.

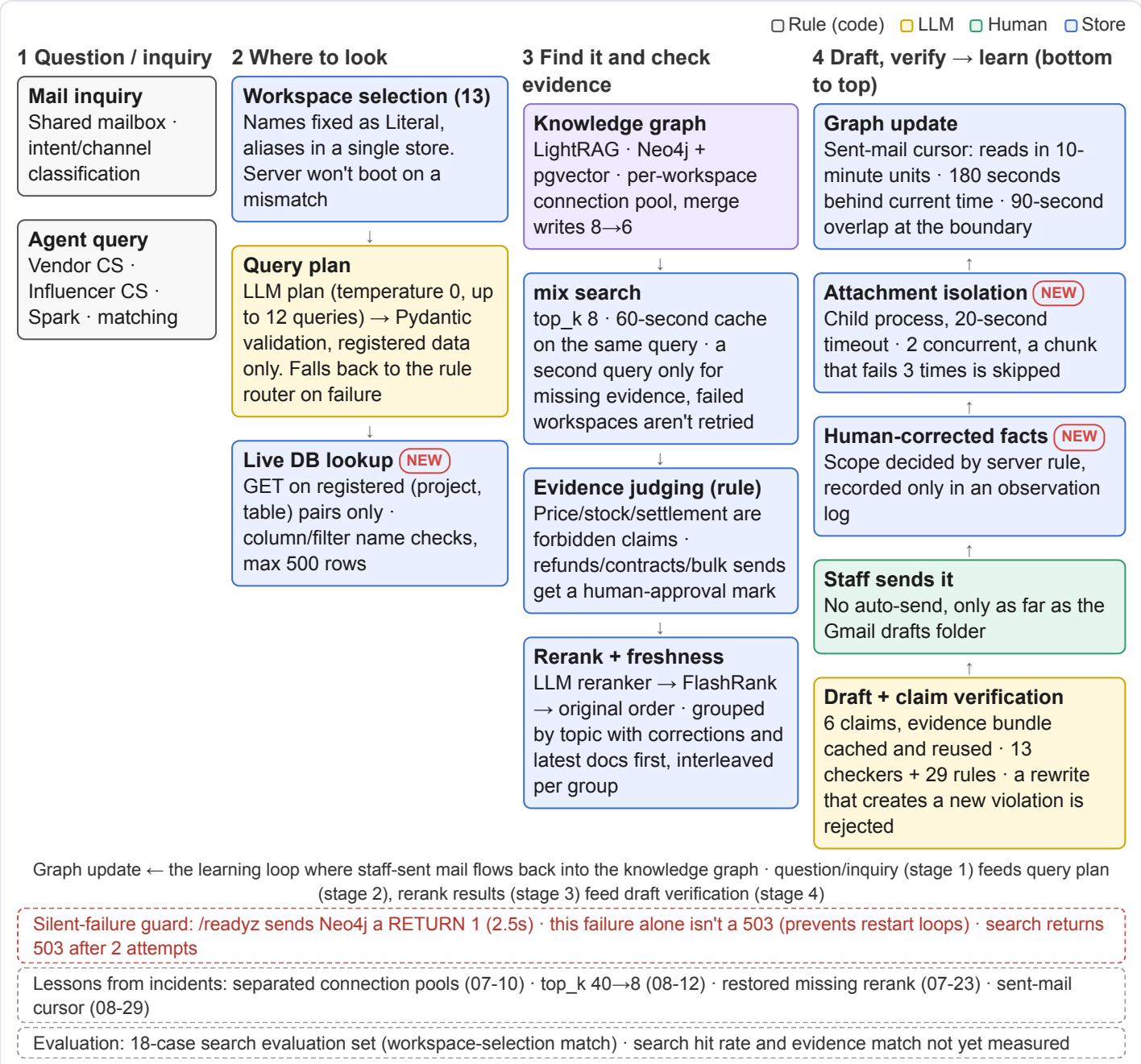
The evidence cache reuses bundles of up to 150,000 characters for five minutes, with a fallback to the original path. Reranking considers up to 20 documents; search concurrency is limited to 3.

> The search evaluation has 18 cases, but currently checks workspace selection rather than evidence hit rate. The effect of restoring reranking has not been measured. Sent-mail learning uses overlapping windows and a cursor behind current time to account for Gmail indexing lag. If I did it again, I would add evidence-match scoring to the evaluation set first; it would have caught the missing rerank before draft quality dropped.

LightRAG Knowledge Hub · Email Agent

Design and operations

Architecture



Technical highlights

1. Validate workspace names at startup. Thirteen canonical names and about fifty aliases resolve to fixed stores. Missing entries or mismatches halt startup.

2. Validate the search plan. Only registered table pairs, GET requests and up to 500 rows pass; invalid plans fall back to the rule router with a recorded reason.

3. Keep reranking available. The fallback sequence is LLM reranker, FlashRank, then original retrieval order. Search concurrency is bounded to reduce graph load.

4. Check distinct failure patterns. Thirteen checkers, each with a real-email replay case, catch reversed requests, invented prior messages and proposals treated as final. Regex gave false positives, so LLM checkers now judge.

5. A rewrite cannot add a new falsehood. Reject a newly introduced factual violation even if the total score improves. Drafts with unresolved violations are discarded.

6. Reuse evidence and learn cautiously. A five-minute cache reuses evidence bundles; overlapping sent-mail windows cover Gmail lag; staff corrections stay in an observation log until their scope is verified.

Work

Internal Work Tool

StyleSeller · Private repositories

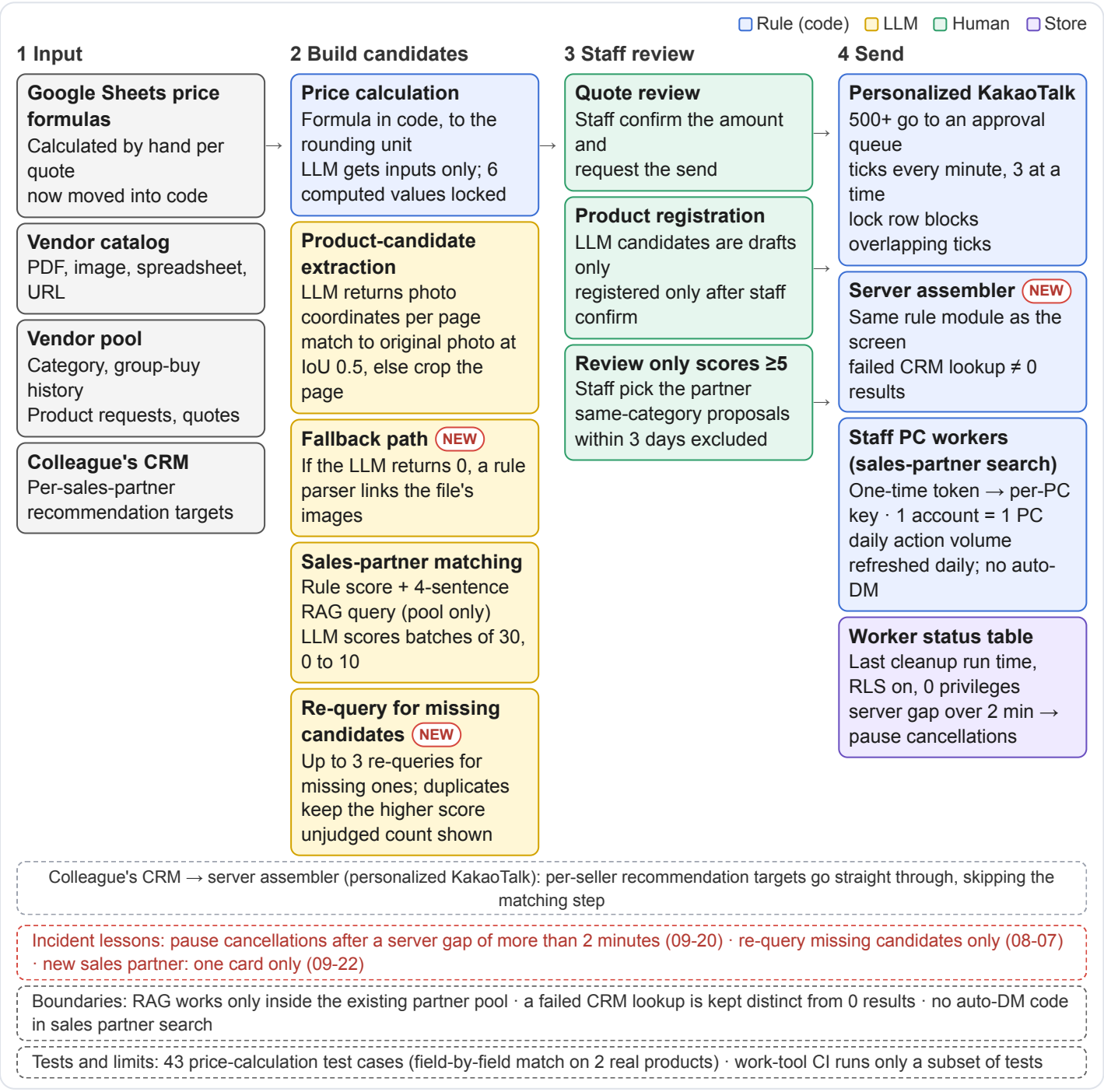
Reliable pricing, reviewable catalog candidates and controlled personalized KakaoTalk sends.

Period	Late March 2026 onward · In production
Role	I built pricing, catalog extraction, partner matching, send automation and PC workers; a colleague's CRM supplies the seller-specific recipients my automation sends to. I used AI coding tools; I set the scope and verified results.
Stack	React 19.2 · Tailwind 4 · Vitest 4 · Postgres + PostgREST · Gemini Flash (LiteLLM) · knowledge-hub RAG · kakaocli · staff PC workers

Background

Pricing depended on spreadsheet formulas and manual checks. I moved the calculations into code and compared real-product values field by field. The tool also extracts catalog candidates and finds partners; staff register and review.

Architecture



Internal Work Tool

> **Scale and evidence.** Personalized KakaoTalk sent to 261 people in one day (2026-09-22) · 43 pricing test cases, some compared field by field against sheet values for real products · Seller-finder worker on 9 staff PCs (2026-09-18)

How it works

1. Receive inputs: Spreadsheet rules, supplier PDFs/images/sheets/URLs, the existing partner pool and a colleague's CRM provide the inputs.
2. Calculate and propose: Code computes prices. The LLM returns editable inputs and product candidates. Rules and RAG widen the partner candidate set within the registered pool.
3. Review: Staff confirm prices and catalog items. Matching excludes recent same-category proposals. Sends to 500 or more recipients wait for approval.
4. Run scheduled work: The server assembles sends in small batches. Workers use per-device keys and account assignment. Seller-finder workers have daily limits and contain no auto-DM code.

Incidents and fixes

Missing match candidates (2026-07-15). In 3 of 6 chunks (the batches sent to the LLM separately), the LLM ended its response before judging all candidates. Since 07-16, candidates are collected as a union and then split for querying, and since 08-07 only the missing candidates are re-queried, up to 3 times. The count that still goes unjudged is shown separately in the UI.

All worker PCs shared one key (2026-09-10). Every staff PC worker was sharing a single key. I changed it so each PC gets its own key through a 30-minute one-time install token. The server stores only the key's sha256, and comparison uses timingSafeEqual.

Six runs cancelled at once (2026-09-20). When the server came back after being down for two days, 6 in-progress runs were cancelled at once. The 'worker not seen for 3 minutes' rule read the server gap as workers dropping out. If the interval between cleanup runs exceeds 2 minutes, it is now treated as a server gap and the reference time is reset.

Technical highlights

1. The LLM edits inputs, never computed amounts. A request such as "set seller profit to 1000" would make the LLM back-calculate and shift other amounts, so six calculated values are blocked from edits. A margin-only request changes the relevant input; code recalculates prices.

2. Match spreadsheet rounding exactly. 43 tests cover two real-product comparisons (including a KRW 30,000 sample). Both shares sum to total profit, which stays unchanged when their ratio changes.

3. Prefer original product images. Match LLM photo coordinates to embedded PDF images at overlap above 0.5. Exclude backgrounds covering over 80% of the page; crop only if no original matches.

4. Recover from incomplete extraction. A rule-based parser handles empty LLM results and ignores icons below 750 bytes. Missing match scores are retried up to three times; unresolved counts stay visible.

5. RAG only widens the existing pool. Rules and search candidates are combined, but unknown partner IDs are rejected. I moved this logic from n8n into code so vitest can test it. The LLM scores batches of thirty; staff see scores of five or more.

6. Manual and scheduled sends share rules. Both use the same product-selection and template code. CRM failure differs from an empty result; a new-seller card is generated once rather than per recipient.

7. Scheduled work stays bounded. Ticks assemble small batches, skip when a backlog exists and stop starting outside the send window. Locks prevent overlap; each account is assigned to one PC.

Design choices

Code calculates amounts; the LLM proposes input edits. RAG widens the candidate set only within the existing partner pool. Sends to 500 or more recipients wait in an approval queue.

PCs Staff PC Deployment

StyleSeller · Private repositories

Shipping and operating tools on staff Macs and Windows PCs, then staying responsible after installation.

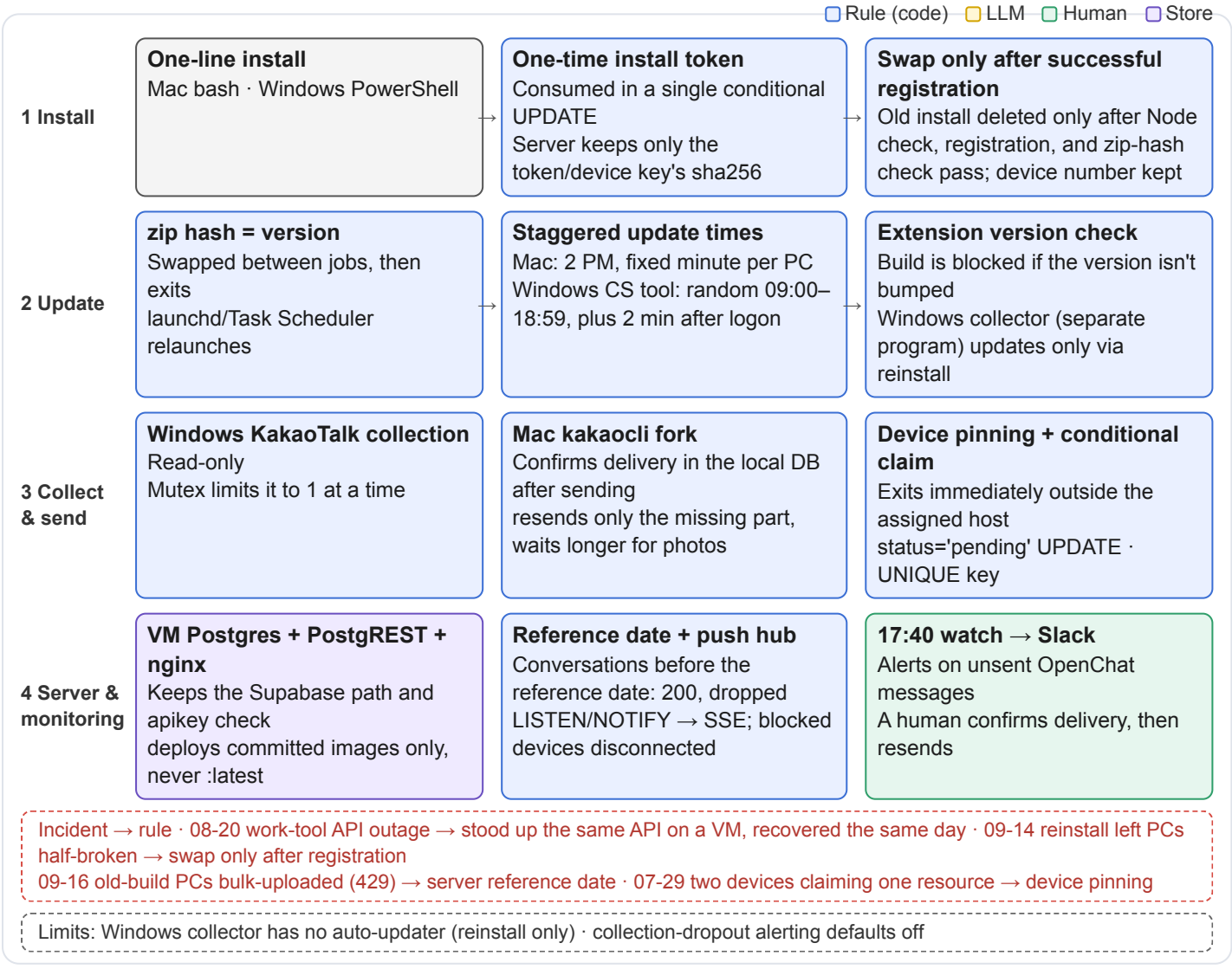
Period	February 2026 onward (Instagram DM extension from February, KakaoTalk tools from May) · In production
Role	I designed installation, authentication and update behavior, built installers and workers, improved the kakaocli fork and Windows collector, and handled releases, monitoring and incident recovery. I used AI coding tools; I set the scope and verified results.
Stack	macOS bash and Windows PowerShell installers · launchd/Task Scheduler · Node workers · Python collector · kakaocli fork (Swift) · Postgres + PostgREST + nginx on a VM · SSE push hub · Docker deploy scripts · Slack watch cron

Background

These workflows cannot rely on a server-only integration with staff KakaoTalk and Instagram sessions. I built local workers, one-line installers and device authentication, then operated the supporting servers. The deployment snapshot recorded eleven staff PCs, with seller-finder workers on nine PCs.

> **Scale and evidence.** 11 staff PCs (as registered on 2026-09-16), 9 seller-finder worker PCs (09-18) · 4 DBs moved to self-hosting (2026-08-06~08-25) · Full work-tool API outage restored the same day (2026-08-20)

Architecture



Staff PC Deployment

How it works

1. Install and register: A one-line command validates the runtime and exchanges an expiring, single-use token for a device key. The server stores SHA-256 hashes.
2. Update safely: The package hash identifies the build. Workers update between jobs and are restarted by the OS scheduler. Update times are staggered by PC.
3. Collect and execute: Jobs are assigned to a device and claimed with one conditional database update. Windows collection is read-only. Mac sends are confirmed in the local database.
4. Observe and recover: A VM serves the API and push hub. Key changes revoke device access; Slack monitoring flags unsent work. Deployments use images tied to a commit.

Incidents and fixes

Full work-tool API outage (2026-08-20). The cloud DB hit its egress quota and every work-tool API stopped. I had nginx on the VM mimic the Supabase paths and apikey header check exactly. The apps changed only their connection settings and a few adapters; the 100-odd createClient calls in app code stayed as they were, and service was back that day.

Reinstall broke healthy PCs (2026-09-14). The installer deleted the old install first and registered after. When it ran with an expired code, the PC was left with no config and no package. Now the old folder is deleted only after the Node check, registration, and zip hash check all pass. The device number stays the same across reinstalls.

Old-version PCs uploaded their whole chat history (2026-09-16). Three Windows PCs sent their backlog of chats all at once. The Sheets API returned rate-limit errors (HTTP 429), and rows for the 2 staff who installed later stayed empty for over an hour. Fixing the client does nothing until the PC picks up the new version. So the server holds a cutoff date, and messages before it are accepted with 200 and then dropped.

Technical highlights

1. Consume install tokens atomically. One conditional UPDATE checks expiry and marks the token used. Two PCs cannot register with the same token; each device has a separate key, so a compromised device revokes only its own row. Failure reasons are written for staff.

2. Use the package hash as the version. Workers swap packages only between jobs and exit; launchd or Task Scheduler restarts them, so no version ledger drifts and two workers never break the Instagram profile.

3. Stagger releases by PC. Different update times leave a window to withdraw a bad release. Mac tools and the Windows CS tool auto-update; the separate Windows collector requires reinstalling.

4. Confirm sends in the local database. After a photo-only send was marked successful (2026-09-07), I added checks for text and photo arrival and resend only missing parts.

5. Protect device assignment and claims. The most frequent incident: two devices taking one resource. On 2026-07-29 a dev Mac claimed the remote-only Mac’s queue; it took a day to trace. Unassigned workers now exit; conditional claims and a unique send-date/room key block competing work; Windows collection stays read-only.

6. Push revocations before more work. Postgres notifications wake workers over SSE. Key changes trigger re-authentication; a revoked device's connection closes.

7. Deploy the script and image from one commit. The 2026-09-28 deploy failed because the running script differed from the commit’s. deploy.sh now re-runs from the target commit, sha-<commit> images never fall back to :latest, and compose renders before the 40-minute build to catch missing variables.

> Counts are dated: eleven PCs registered on September 16 and nine seller-finder workers on September 18. Mac tools and the Windows CS tool update automatically; the separate Windows collector still requires reinstalling. Collection-gap alerts are off by default. A missing release image stops deployment rather than selecting another version.

Next time I would ship the Windows collector with a zip-hash updater from day one; the 09-16 incident grew because old PCs missed the client fix. Messages with unreadable timestamps are kept, not dropped. Recovery preserves the API contract, so client changes stay limited to connection settings and a few adapters.